

Raj Aaryan

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Skills

- **Languages:** Java, C++, Python, JavaScript
- **Frameworks & Technologies:** Spring Boot, Kafka, REST APIs, Camunda, Microservices, Kubernetes, Elasticsearch, Redis, JUnit, Mockito
- **Databases:** MySQL, DynamoDB
- **Concepts & Tools:** System Design (LLD/HLD), OOP, Design Patterns, SOLID Principles, Data Structures & Algorithms, Git, CI/CD

Work Experience

Jio Platforms Limited

Jul 2022 – Present

Software Development Engineer

- Developed and maintained 3 core microservices (COM, SOM, ROM) for Jio's Digital API Transformation Platform using Java 17 and Spring Boot, orchestrating end-to-end order lifecycle management and integrating with 20+ downstream services following TM Forum Open API standards.
- Spearheaded automation of the order orchestration pipeline using Camunda BPM workflows and BPMN process definitions, reducing manual intervention by 40% and order processing time by 97%.
- Built a fallout management framework to proactively detect, track, and auto-resolve upstream failures, improving system reliability and reducing processing failures by 35%.
- Designed and developed RESTful APIs using Spring MVC, Spring Data JPA, Hibernate, and OpenAPI 3.0 standards, supporting large-scale order processing workflows.
- Integrated Kafka for asynchronous inter-service communication, decoupling order processing from downstream dependencies and improving resilience during peak traffic conditions.
- Implemented centralized logging, monitoring, and observability using Elasticsearch, Kibana, and Kubernetes across 10+ microservices, reducing debugging effort and mean time to detection by 30%.
- Built reporting and analytics APIs using MariaDB JSON functions, exposing 50+ configurable fields and reducing manual data extraction effort by 80%.

Samsung Electronics

Aug 2021 – Jun 2022

Software Engineer Intern

- Developed embedded firmware modules in C++ for NAND Flash memory devices, improving storage reliability and data integrity.
- Enhanced Bad Block Management (BBM) logic for NAND Flash memory devices, reducing bad-block handling latency by 20% and improving storage reliability.
- Optimized encryption and decryption modules in C++, reducing encryption overhead by 15% while maintaining compliance with security requirements.

Projects

E-commerce Order Processing System

Java, Spring Boot, Kafka, Redis, MySQL [GitHub]

- Designed a distributed e-commerce order processing platform using Spring Boot microservices for Order, Inventory, Payment, and Shipment workflows.
- Implemented Saga-based orchestration and Kafka-driven event processing with retry and DLQ mechanisms to ensure eventual consistency and fault tolerance.
- Leveraged Redis caching and idempotent APIs to reduce latency and prevent duplicate order processing.

Distributed Rate Limiter – Low Level Design

Java, Redis, Microservices [GitHub]

- Designed a distributed rate limiter using Token Bucket and Sliding Window algorithms for high-throughput APIs.
- Implemented Redis-based atomic counters and expiry mechanisms to ensure consistency across distributed instances.

Education

BITS Pilani (Pilani Campus)

2017 – 2022

Dual Degree: B.E. in Electronics & Instrumentation, M.Sc. in Mathematics

Achievements

- Received MCN Scholarship for 10 consecutive semesters for academic excellence.
- Received A* rating (top 1% performer) for three consecutive years at Jio Platforms.
- Mentored an intern on system design and onboarding, contributing to team productivity.